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Protocol

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Fabrication and characterization of optoelectronic in-sensor computing devices

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Abstract

Bioinspired in-sensor computing devices can process information at sensory terminals by leveraging physical principles, thereby reducing latency and energy consumption during computation while simultaneously enhancing the efficiency of data processing and real-time analysis. Optoelectronic devices exhibit in-sensor computing functions, such as feature enhancement and data compression, by tuning the defect states of the semiconductor channels and thereby modulating the photoresponsivity and time constants of the sensors. These functionalities are critically dependent on precise fabrication and testing protocols. Here we present a detailed procedure for fabricating and characterizing in-sensor computing devices based on nanoscale semiconductor thin films. We explain how to test such optoelectronic devices, including the testing of visual adaptation and motion perception responses. When using semiconductor materials obtained from commercial suppliers, this procedure is time efficient and results in highly reproducible device performance. Nevertheless, all device fabrication and testing steps are generalizable and can be extended to other semiconductor thin films grown using different methods. The procedure is intended for researchers experienced in cleanroom operations and microfabrication techniques and can be completed in ~14 d. The use of bioinspired optoelectronic devices enables the development of a framework for advancing in-sensor computing technologies.

Key points

- The first section of the procedure details the fabrication of sensors with features for the control of interface defects and quality, which enable the trapping and detrapping of carriers necessary for in-sensor computation.
- The second section of the procedure details optoelectronic characterizations of the device, including electrical characterization, optoelectronic response characterization and bioinspired in-sensor computing characterization.

Key references

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Introduction

The data generated by Internet of Things devices is experiencing exponential growth, driven by the proliferation of connected sensors[1–3]. This surge in data volume presents grand challenges in data management, transmission, storage and analysis and increases the demand for information processing close to the sensory terminals[4–6]. Computing with physics of sensory devices can alleviate the strain on data transmission between sensors and computation units[7]. Vision is the dominant information source among sensors due to its ability to capture rich, detailed data from the environment[8]. Machine vision systems capture abundant information under a variety of illumination conditions[4,9,10], which puts considerable strain on computational resources and complex hardware systems to reconstruct visual information[1,11–13]. Their further development requires robust and reliable hardware with high image-capturing speed[1,2], high spatial resolution[14] and a high sensitivity under various lighting conditions[15]. Although state-of-the-art vision sensors have seen much improvement[8,9,16–20], they still lack computational functions[4,16,21].

Biological visual systems can perceive and recognize objects in a dynamically changing and complicated environment with high efficiency and agility[4,22,23]. The abundant visual information is processed within the visual system, thereby reducing the transmission of excessive visual data to the brain[5]. These evolutionary features may be useful for the design of in-sensor computing devices that can adapt to varying light intensities and different light wavelengths[4,17] and perceive and recognize the dynamics of motion[5,16]. We aim to reproduce these features in integrated in-sensor computing devices. The dynamic range of traditional complementary metal–oxide–semiconductor (CMOS) sensors is usually 70 dB, which is much narrower than the light intensity range across natural scenes[4]. This limited dynamic range leads to the loss of visual information in natural environments with widely varying illumination intensities. In comparison, optoelectronic in-sensor computing devices can effectively perceive up to 199 dB (ref. 4), which can improve the accuracy of the scene perception. Traditional CMOS sensor architectures also separate sensing from computing units and require massive data transfer between memory and processing modules, leading to significant power consumption[2]. The segregated processing in traditional CMOS sensors further implies that collection of temporal and spatial information needs to occur independently and subsequently fused for motion perception, inevitably requiring additional power consumption[5]. In comparison, optoelectronic in-sensor computing devices can directly sense and process information, effectively encoding both temporal and spatial information streams at the sensory terminals, thereby reducing redundant visual data transmission and subsequent computational resources and reducing power consumption. The photoresponsivity of the vision sensors can be dynamically modulated by charge trap states in photoactive materials under different illumination conditions, enabling efficient information processing. To broaden their applications, these sensors need to be straightforward to fabricate and characterize.

Here, we provide detailed procedures for the fabrication and characterization of optoelectronic in-sensor computing devices and complement them with a few example applications. We use two-dimensional (2D) layered semiconductors to construct the optoelectronic in-sensor computing devices because of their strong light–matter interactions[24–27], effective electrostatic modulation[28–31] and distinctive physical properties[32–35]. The device physics can be also extended to other semiconductor materials[6,10,12,36]. Based on our team's previous research on in-sensor visual adaptation and motion perception[4,5,8,16,17,33,37], we demonstrate the in-sensor computing functions, including visual adaptation and motion perception.

Development of the protocol

In recent years, various materials have been utilized in the design and fabrication of optoelectronic in-sensor computing devices, including 2D transition metal dichalcogenides[1,4,5,16,33,38–40], intrinsic silicon[6], photochemical reaction materials[12,37] and other materials suitable for photocurrent modulation[36,41–45]. Optoelectronic in-sensor computing devices based on such materials are mainly divided into optoelectronic nonvolatile or volatile

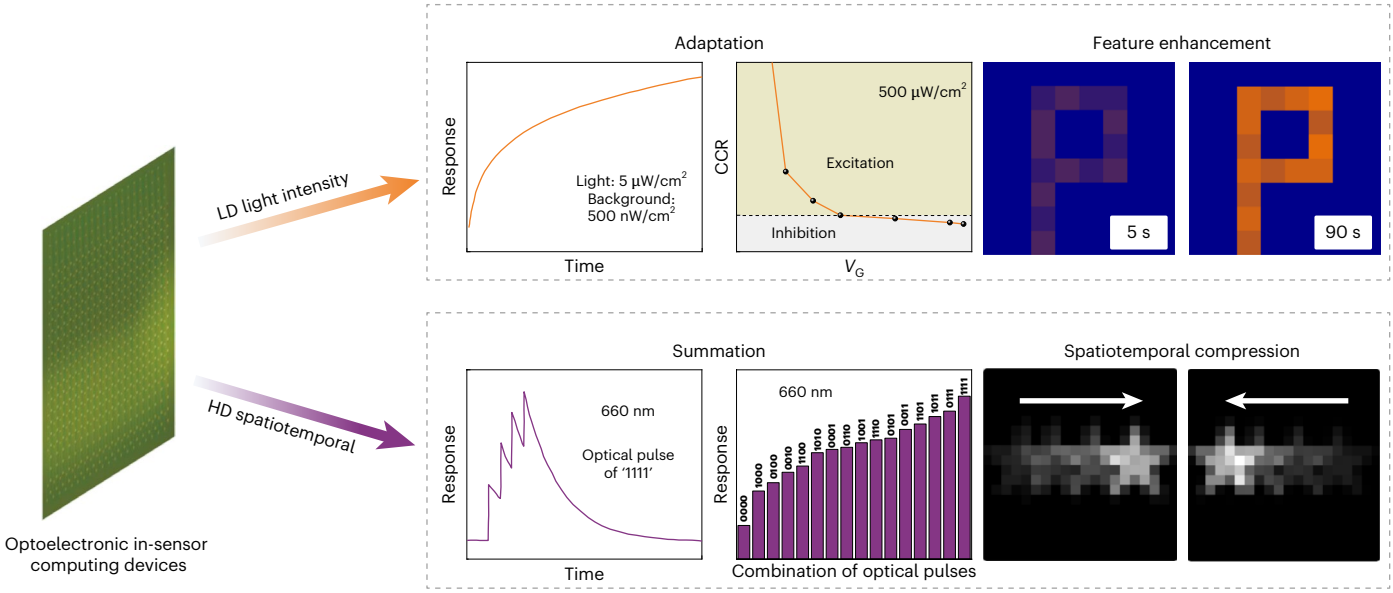


Fig. 1 | Functionality and application demonstration of optoelectronic in-sensor computing phototransistor array. For LD light intensity perception, optoelectronic in-sensor computing devices enhance features by establishing a functional relationship between input light intensity and output light response. For HD spatiotemporal perception, in-sensor computing devices effectively fuse spatiotemporal visual information by the summation of light response over time to achieve data compression.

devices, which usually modulate carriers through different mechanisms[37]. These optoelectronic devices can directly process raw visual information in both spatial and temporal dimensions[37]. In the spatial dimension, the in-sensor computing devices can realize feature enhancement by establishing a functional relationship between the input light intensity and the output light response[8,37]. In the temporal dimension, optoelectronic in-sensor computing devices can effectively process dynamic visual information and enable data compression by summing the current and previous light responses over time[8,37]. We developed bioinspired in-sensor computing phototransistors based on MoS_2 and applied them to enhance feature extraction in static visual scenes. These sensors can mimic the adaptative functions of the human retina, allowing time-varying light responses for visual adaptation (Fig. 1). To improve visual adaptation, enhancements in phototransistors are needed for image contrast and dynamic range[46–48]. To achieve this, we intentionally introduced more charge-trapping states on the MoS_2 surface, as these defect states can effectively affect the optoelectronic characteristics of in-sensor computing devices under various light stimulation by trapping or detrapping electrons in the channel under different gate voltages. Therefore, this protocol features the ultraviolet ozone (UVO) treatment of MoS_2 films to generate a large number of trapping states and tests the optoelectronic performance and visual adaptation effect of the fabricated in-sensor phototransistors.

In terms of functionality and perception dimension, in-sensor computing devices can respond to the spatial distribution of light intensity, which allows the enhancement of the features detectable in low-dimensional (LD) static scenarios. However, real-world visual environments are dynamic and complex. It is very important to develop high-dimensional (HD) in-sensor computing devices that can effectively perceive motion evolving in the temporal domain. We developed in-sensor computing phototransistors that can mimic the graded neurons of the flying insect visual system, enabling the perception of various types of motion that fuse temporal and spatial information[5] (Fig. 1). The time constant of these sensors can impact temporal evolution and summation of the photocurrent through gate voltage modulation[8,37], facilitating spatiotemporal data compression and reducing data volume when perceiving dynamic scenarios. Taking the pentagram pattern as an example, the MoS_2 phototransistor arrays allow the perception of spatiotemporal vision information (Fig. 1).

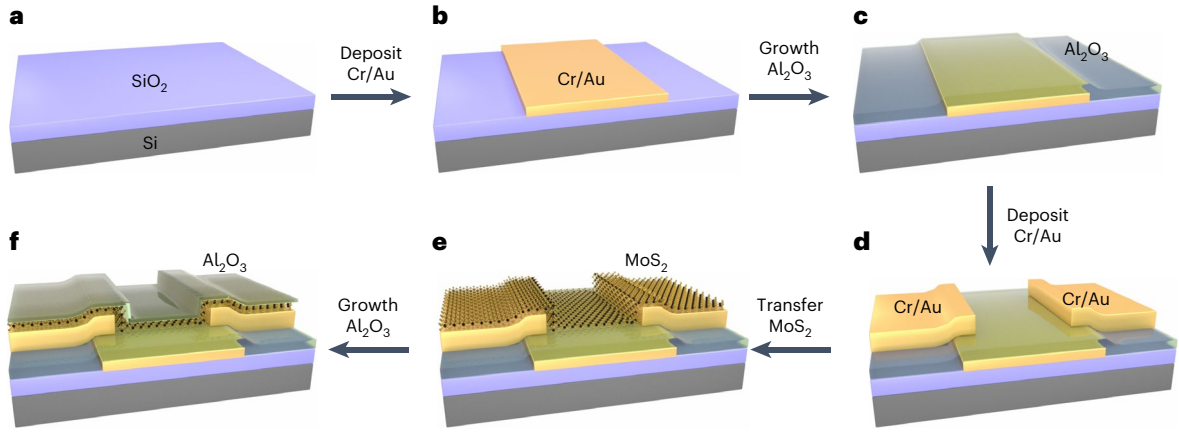


Fig. 2 | Schematic of the fabrication process flow. a, A schematic of cleaning the Si/SiO_2 substrate. **b**, A schematic of depositing a back-gate electrode (Cr/Au). **c**, A schematic of depositing gate dielectric (Al_2O_3). **d**, A schematic of depositing source and drain electrodes. **e**, A schematic of transferring 2D semiconductors (MoS_2). **f**, A schematic of preparing a passivation layer.

The quality of semiconductor materials plays a crucial role in the performance of optoelectronic devices for in-sensor computing functionalities. In this procedure, we use large-area MoS_2 thin films from various channels, including chemical vapor deposition (CVD) and metal–organic CVD (MOCVD) techniques (see 'Materials' section for more details). This approach allows us to develop multifunctional and large-scale optoelectronic devices with excellent uniformity and high yield. In addition, these devices can be efficiently prepared using easily available, off-the-shelf tools, making this approach highly valuable for the large-scale fabrication of bioinspired in-sensor computing devices. The testing method for the in-sensor phototransistor array described in this protocol can be adapted to various types of device that integrate sensing and processing, offering a practical approach for assessing large-scale systems with similar functionalities.

Overview of the procedure

Considering the importance of both fabrication and measurements for optoelectronic in-sensor computing devices, we divided the procedure into two key parts: fabrication and testing. As shown in the process flow diagram (Fig. 2), the fabrication consists of six main stages, divided into 33 steps. In Steps 1–2, we describe the cleaning process for the Si/SiO_2 substrate, which is essential for enhancing optoelectronic performance of the devices. We then present the detailed steps of fabricating the back-gate metal electrode and marker (Steps 3–5). After the atomic layer deposition (ALD) growth of the gate dielectric (Steps 6–7), the source and drain metal electrodes are fabricated using a process similar to that for the back-gate electrode (Step 8). In Steps 9–20, we describe the most important yet challenging process of the procedure: wet transfer and etching of 2D semiconductor materials, which provides the material basis for optoelectronic in-sensor computing devices. Supplementary Videos 1 and 2 provide comprehensive visual assistance. After preparing the channel layer, we describe the procedure for the annealing and UVO treatment of the devices (Steps 21–22), which can improve the contact and promote the introduction of defects, respectively. To ensure that the 2D devices maintain long-term stability in air, the preparation and etching of the encapsulation layer are described in detail in Steps 23–25. These processes enable the control over interface defects and quality of in-sensor computing devices, which in turn affect the trapping and detrapping of carriers, thereby influencing feature enhancement and data compression of in-sensor computing devices. In Steps 26–33, we describe how to initially verify the quality of the device, including photography, scanning electron microscopy (SEM) image, Raman spectra and photoluminescence spectroscopy (PL).

The testing protocol also consists of five main stages, divided into 24 steps. In Steps 34–39, we describe basic electrical characterizations of optoelectronic in-sensor computing devices,

including transfer and output curve measurement, charge defect characterization and uniformity statistics. Then, we provide detailed protocols to characterize the fabricated devices (Steps 40–45), such as light-intensity-dependent characteristics, time-dependent dynamic responses and wavelength-dependent properties. To illustrate the procedure of perceiving light intensity and enhancing features in the spatial dimension, we demonstrate the characterization of perception range (PR) modulated by gate voltage and the adaptation characterization of the output photocurrent over time, as explained in Steps 46–49. Next, we detail how bioinspired in-sensor phototransistor arrays can enhance image contrast and enable accurate and efficient visual adaptation (Steps 50–52). Compared with devices designed for perceiving LD light intensity and static scenarios, we have also developed devices capable of HD spatiotemporal and motion scenarios perception, which effectively fuse spatiotemporal visual information and enable data compression by the summation of light response over time. The graded neurons characterizations of devices are demonstrated in detail to show the photocurrent summation characteristics (Steps 53–56). We then detail the implementation of bioinspired phototransistor arrays for accurate and efficient motion perception as well as spatiotemporal data compression (Steps 57–58).

In-sensor computing applications

The optoelectronic in-sensor computing device enhances speed, energy efficiency, architectural compactness (that is, a simplification of the hardware) and can be used under varying illumination conditions[4]. In complex environments, the illumination intensity often exceeds the effective range of some visual sensors and can lead to the loss of critical information. An adaptive function enables the visual sensor to dynamically adjust its photoresponsivity on the basis of the background illumination intensity, thereby preserving sensitivity to context-relevant information. Under dim light conditions, the devices can exhibit increased sensitivity over time to clearly detect objects, made possible by the gradual release of electrons from shallow and deeper traps of the channel material after applying a negative gate voltage (V_G)[4], resulting in an increase in the photocurrent of optoelectronic in-sensor computing device. This effect is similar to the adaptation of the human eye to changes in dim light environments. Under continuous strong light conditions, the devices can adjust the channel material by applying a positive V_G, causing shallow and deeper trap states to gradually trap electrons over time[4]. This process reduces the sensitivity of devices to clearly perceive objects, in an analogous manner used to adapt the human eye to high-intensity illumination environments. Thus, optoelectronic in-sensor computing devices with adaptive functionality can deliver accurate results across varying illumination conditions.

The optoelectronic in-sensor computing devices can also be used for real-time efficient motion perception. They can simultaneously encode temporal and spatial information at sensory terminals by modulating the shallow charge-trapping centers of channel materials with V_G, thereby accurately perceiving the motion scenes and effectively reducing computing resources[5]. In addition, the V_G of devices can dynamically modulate the time constant of charge-trapping centers from 10^1 to 10^6 ms, which allows them to perceive a wide range of motion scenarios, from slow movement to fast running.

From a broader perspective, optoelectronic in-sensor computing devices have been used in event-driven sensing[16,49], spectral adaptation[17], polarization detection and other fields[8,37,50]. Moreover, the device-preparation process and testing methods can be adapted for other 2D materials, paving the way for realizing 2D vision electronics.

Limitations

Although optoelectronic in-sensor computing devices based on 2D MoS_2 films demonstrate efficient visual perception, their fabrication faces several limitations. While 2D layered materials exhibit excellent flexibility and can withstand considerable strain[51–53], wet-transferred centimeter-scale MoS_2 films onto back-gate substrates are often susceptible to cracks and flaws due to the tension of solution or thermal process. We have found that using thermal release tape-assisted transfer can help to mitigate this problem (Steps 9–15); nevertheless, it remains

challenging to eliminate cracks and flaws entirely. A critical engineering task is to enable the transfer or growth of high-quality, large-area MoS_2 films onto the specific gate dielectric layer without wrinkles and cracks. Another essential problem that needs to be solved is minimizing contamination during the device-fabrication process. These contaminants, which may originate from photoresist and other chemicals, can introduce unwanted polymer residues into the channels, thereby degrading the device performance. Therefore, careful and comprehensive wash steps are necessary in the whole fabrication process.

For visual perception testing, the spatial resolution limitation is unneglectable based on the prepared phototransistor arrays. The relatively low density of the sensors results in limited spatial resolution. In the future, optimizing the array layout design and reducing the electrode area can increase pixel density per unit area, which in turn enhances spatial resolution. In addition, a major challenge in testing phototransistor arrays is that the probes are used to test each device point by point, which is highly time-consuming and significantly impedes practical applications. To overcome this problem, it is essential to design matching readout circuits, enabling rapid testing of phototransistor arrays. Moreover, it is highly challenging for simple phototransistors to exhibit visual adaptation. Achieving self-adaptive visual functionality in phototransistors fundamentally requires that the devices autonomously modulate their operating state according to ambient light levels, enabling passive transitions between scotopic and photopic adaptation. There are two potential approaches to endow phototransistors with visual adaptation functionality. Fabricating floating-gate phototransistors represents a promising approach. By modulating the trap states in MoS_2 via the floating gate, the phototransistor can autonomously modulate its threshold voltage in response to illumination over time, potentially enabling adaptive functionality. The integration of a photodiode with a phototransistor presents another potential approach. The photogenerated electrons produced by the photodiode serve as the gate voltage for the phototransistor. Under strong and weak light conditions, the photodiode generates more or fewer photogenerated electrons, respectively, resulting in a difference in the gate voltage applied to the phototransistor.

Comparison with other methods

The optoelectronic in-sensor computing devices fabricated using this protocol have several advantages over state-of-the-art image sensors based on CMOS technology. Accurate image capture under varying illumination conditions is essential for reliable environmental perception, as natural light intensity spans an extremely wide dynamic range of ~280 dB (ref. 4). The dynamic range of state-of-the-art silicon CMOS image sensors is usually 70 dB, which may lead to the loss of visual information with widely varying illumination intensities[4]. In comparison, the fabricated devices can effectively perceive up to 199 dB (ref. 4), which can achieve more accurate scene perception in both bright and dim light conditions. In addition, for motion perception, conventional CMOS image sensors first collect static frames. The multiple spatial frames form a temporal video, further processed by a classical neural network with a greater number of layers and network parameters for action recognition. The frame-to-video progress results in low transmission rate for the sensory system. Further, the perception of motion information often relies on the results of collecting and postfusing timestamps and spatial stamps separately, which can lead to increased power consumption. In comparison, the devices can mimic the visual system of flying insects and effectively encode both 'temporal' and 'spatial' stream information at sensory terminals, thereby reducing redundant visual data transmission and subsequent computational resources, which can reduce power consumption.

Another important advantage of this method is the tunability with the gate voltage. Compared with commercial photodiodes[18,54], the phototransistors can amplify the photocurrent through gate voltage modulation, resulting in significantly enhanced photoresponsivity. As a result, the fabricated devices can achieve visual perception under weaker light conditions. Moreover, the gate voltage can also adjust the falling time of the photoresponse, thereby achieving dynamic perception across a wide range of motion speeds.

Materials

Reagents

▲ **CRITICAL** MoS_2 materials must be stored in an argon-filled glove box to prevent deliquescence in air. In addition, containers holding MoS_2 devices or materials should be wrapped by aluminum foil to shield them from light.

- Molybdenum disulfide (monolayer) (MoS_2; synthesized by CVD, SixCarbon Technology Shenzhen)
- MoS_2 (bilayer) (synthesized by MOCVD)
- 4-inch silicon substrates with 300-nm-thick silicon oxide layer (one-side polished, type/orientation P<100>, ~500-μm thickness, 300 nm SiO_2) (KJ Group, cat. no. KMT110NMSIO2)
- Alumina, 50 nm (Al_2O_3; synthesized by ALD, Laajoo, cat. no. FN0002)
- Poly(methylmethacrylate) (PMMA; 950 K, Kayaku Advanced Materials, cat. no. 22090614)
  ▲ **CAUTION** This chemical is a flammable liquid and vapor that can be harmful if inhaled. It can cause skin irritation and severe eye irritation and may also cause respiratory irritation. To ensure safety, it needs to be handled in a ventilated fume hood. Proper protective equipment, such as a lab coat, gloves and safety goggles, is strongly recommended during use.
- Potassium hydroxide (KOH; Aladdin, cat. no. A2415491)
  ▲ **CAUTION** This chemical is a water-soluble solid. It is nonflammable but dissolves easily in water, releasing heat and forming a corrosive solution. Inhalation can cause severe respiratory irritation or burns. Direct contact with skin or eyes may result in burns. It is strongly recommended to wear appropriate protective equipment while handling.
- Microposit AZ5214E Photoresist (MicroChemicals, cat. no. USAW543501)
  ▲ **CAUTION** This chemical is a flammable liquid and vapor. It can cause skin and eye irritation. Inhalation may damage the organs. It should be handled in a well-ventilated area. Specific gloves, goggles and protective clothing must be worn during handling.
- Microposit PMGI SF6 Photoresist (Kayaku Advanced Materials, cat. no. 23120847)
  ▲ **CAUTION** This chemical is a flammable liquid and vapor. It can cause skin irritation and serious eye damage. Inhalation may damage the respiratory system and kidneys. It is strongly recommended to use it in a well-ventilated area and wear appropriate protective equipment.
- Microposit AZ300 MIF Developer (MicroChemicals, cat. no. KR517629)
  ▲ **CAUTION** This chemical can cause skin and eye irritation. Inhalation may damage organs. It should be handled in a well-ventilated environment. Specific gloves, goggles and protective clothing must be worn during handling.
- Photoresist remover (Kayaku Advanced Materials, cat. no. 24100797)
  ▲ **CAUTION** This chemical is a flammable liquid and vapor. It is highly toxic, and prolonged contact may cause skin damage. Short-term contact may result in redness or other symptoms. Appropriate personal protective equipment should be used while handling.
- Acetone (C_3H_6O; Anaqua Global International, cat. no. TAE12Y2403TH)
  ▲ **CAUTION** This chemical is a highly flammable liquid and vapor. Inhalation may cause drowsiness or dizziness and could even lead to serious eye irritation. Appropriate personal protective equipment is strongly recommended during use.
- Isopropyl alcohol (IPA, C_3H_8O; Anaqua Global International, cat. no. TIA82Y2305TH)
  ▲ **CAUTION** This chemical is a highly flammable liquid and vapor. Inhalation may cause drowsiness, dizziness, or respiratory irritation. Appropriate personal protective equipment should be used while handling.
- Ultrapure deionized (DI) water (H_2O; Milli-Q system, Millipore)
- Chromium (Cr) pellet (99.99%) (Kurt J. Lesker, for metal deposition, cat. no. CR39854601)
- Gold (Au) pellet (99.99%) (Kurt J. Lesker, for metal deposition, cat. no. AG59669441)

Protocol

Equipment

- Hot plate (Sawatec, model no. HP-150)
- Tweezer (Trusco, product no. 7647751)
- Spin coater (Sawatec, model no. SM-150)
- Weighing balance (Mettler Toledo, model no. MX 204)
- Thermal release tape (Silicon Wafer, model no. 120–140° C)
- Plasma cleaner (PVA TePla, model no. IoN 3)
- Inductively coupled plasma (ICP; Sentech, model no. SI 500)
- Ultrasonic cleaner (Digital Equipment, model no. 10 L)
- Raman spectrometer (Oxford Instruments, model no. WITec alpha300 R)
- PL spectrometer (Oxford Instruments, model no. WITec alpha300 R)
- Optical microscopy (OM; Leica, model no. DM2700)
- SEM (Hitachi, model no. SU1510)
- Atomic force microscope (AFM; Bruker, model no. Dimension Icon)
- Laser direct writing (TuoTuo Technology, model no. UV Litho-ACA Pro)
- Electron-beam evaporator (E-beam; Denton Vacuum, model no. Denton Explorer)
- ALD (Beneq, model no. TFS 200)
- UVO cleaner (Novascan, model no. PSD Pro Series)
- Rapid thermal annealing processor (Allwin 21, model no. AccuThermo AW410)
- Semiconductor analyzer (Keithley, model no. 4200-SCS)
- Probe station (LakeShore, model no. PS-100)
- Laser (LL-Laser, model no. LLG660-3000 mW)
- Pulse generator (FeelTech, model no. FY 2300)
- Tunable xenon lamp (NBeT, model no. HSX-UV300)
- Optical power meter (Newport, model no. 843-R)

Equipment setup

Optoelectronic testing system

- In this protocol, the optoelectronic characterization, primarily including current–voltage (*I*–*V*) and current–time testing, consists of a semiconductor analyzer, a probe station, various lasers with wavelengths of 450, 520, 660, 808 and 980 nm, optical fibers, a pulse generator and an optical platform. Notably, the light intensity from the laser is calibrated using an optical power meter. The semiconductor analyzer is equipped with a source measure unit (SMU) module to test quasi-direct current electrical characteristics, while the pulse generator can control the laser output pulses. During testing, the gate, drain and source electrodes of the phototransistor are connected to the three probes of the probe station, which are linked to the corresponding SMU channels of the semiconductor analyzer. Lasers of various wavelengths are delivered to the device through optical fibers mounted on the optical platform

Procedure

Back gate and metal marker preparation

● **TIMING 1 d**

1. Use AutoCAD software to design the photolithography patterns for the phototransistor and use a laser direct writing equipment to pattern. Choose the area of the photolithography patterns (~3 cm × 3.2 cm) as a balance between lithography time and the subsequent probe test requirement.
   ▲ **CRITICAL STEP** We can prepare four arrays on a 4-inch wafer. A 0.4-cm spacing between adjacent arrays was selected to ensure that each array could be easily separated using a diamond scriber or laser.
2. Clean the 4-inch Si/SiO_2 wafer ultrasonically with acetone, IPA and DI water, and then dry it using nitrogen gas as specified in Recipe 1 (Table 1).

Table 1 | Process recipes and detailed parameters

Recipe code	Type of recipe	Equipment used	Recipe details	Postprocessing and other information
R1	Substrate cleaning	Ultrasonic cleaner	Power, 100%; frequency, 90 kHz; mode, normal; time, 8 min	Solvents used, acetone and IPA; acetone to remove organic contaminants and IPA to remove acetone and residues
R2	Spin-coating photoresist (PMGI SF6 and AZ5214E)	Spin coater	For PMGI SF6, 500 rpm for 5 s, 3,000 rpm for 50 s; for AZ5214E, 500 rpm for 5 s, 4,000 rpm for 30 s	For PMGI SF6 and AZ5214E, bake at 160 °C and 110 °C for 5 min and 1 min respectively, then put on a cold plate for 30 s
R3	Spin-coating photoresist (PMMA)	Spin coater	500 rpm for 5 s, 2,000 rpm for 40 s	Soft bake for 10 min at 110 °C then put on a cold plate for 30 s
R4	Spin-coating photoresist (AZ5214E)	Spin coater	500 rpm for 5 s, 4,000 rpm for 40 s	Soft bake for 1 min at 110 °C then put on a cold plate for 30 s
R5	Dry etching (MoS_2)	Plasma cleaner	Gas used, O_2; pressure, 0.25 Torr; power, 50 W; time, ~50 s	For etching the non-channel regions of MoS_2 film after transfer
R6	Dry etching (Al_2O_3)	ICP	Gas used, BCl_3 and Ar; pressure, 4 mTorr; power, 300 W; bias, 100 W; time, ~1 min	For etching the Al_2O_3 layer in the source, drain and gate regions

3. Spin-coat PMGI SF6 and AZ5214E positive photoresist uniformly on the top surface of the Si/SiO_2 wafer with using Recipe 2 in Table 1.
4. Use laser direct writing (wavelength, 405 nm; energy, 319 mW/cm^2) to pattern the back-gate metal electrode and marker on the baked samples, then develop with the developer for 25 s, rinse with DI water for 20 s and dry with a nitrogen gun.
5. Use an E-beam evaporator to deposit 3/30 nm of Cr/Au as the back-gate electrode and marker, and use a liftoff process with photoresist remover to achieve the patterned metal electrode and marker.
   ◆ **TROUBLESHOOTING**

Gate dielectric preparation

● **TIMING 3 h**

6. As described in Step 2, clean the 4-inch back-gate substrate ultrasonically using acetone, IPA and DI water, and then dry it with nitrogen gas.
7. Use ALD to grow a 40-nm-thick Al_2O_3 gate dielectric at a growth temperature of 250 °C and a deposition rate of ~0.1 nm/s. Subsequently, divide the 4-inch gate dielectric substrate into four arrays, and select one of them for further preparation.
   ◆ **TROUBLESHOOTING**

Source and drain electrode preparation

● **TIMING 1 d**

8. As described in Steps 1–5, after spin coating the photoresist, we used laser direct writing or photolithography to overlay the source and drain electrodes on the gate dielectric substrate according to the cross marker on the substrate. Then, obtain the source and drain electrodes through a development process, E-beam deposition of Cr/Au and a subsequent liftoff process.
   ▲ **CRITICAL STEP** The accuracy of the overlay was critical for the fabrication of array devices. To produce high-performance phototransistors, we recommend that the cross markers on the source and drain electrodes layout be strictly aligned with those on the back-gate substrate, with an alignment error not exceeding 1 μm.

Wet transfer of 2D semiconductors

● **TIMING 1 d**

9. Use CVD to grow a monolayer MoS_2 film on a 2-inch Si/SiO_2 substrate.
10. Cut out a ~3.5 cm × 3.5 cm region with high-quality MoS_2 film for subsequent wet transfer.
11. Spin-coat PMMA photoresist on the upper surface of the prepared sample using Recipe 3 in Table 1.

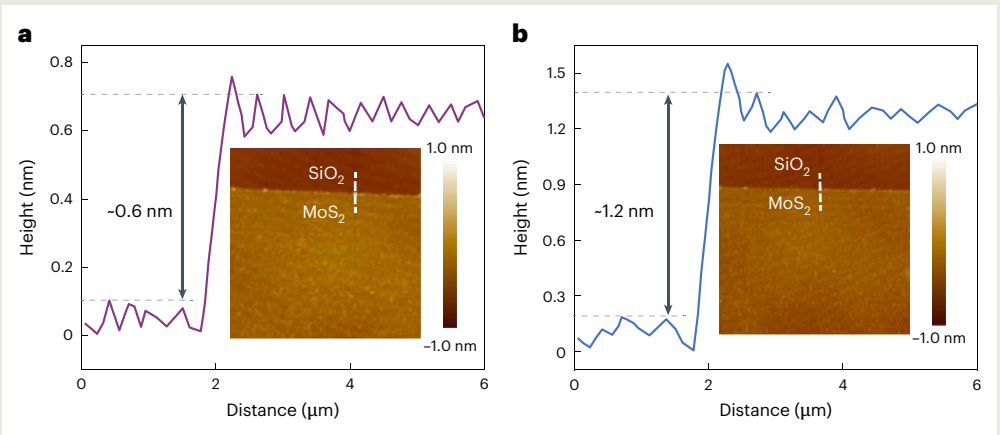
BOX 1

The AFM topography images of monolayer and bilayer MoS_2 films

● **TIMING 2 h**

Procedure

1. Select monolayer and bilayer MoS_2 films grown on Si/SiO_2 substrates for AFM characterization.
2. Obtain the samples for AFM testing by pattern-etching areas and developing and etching the MoS_2 film and removing the photoresist.
3. Place the fabricated samples on the stage of the AFM.
4. Characterize the topography images of monolayer and bilayer MoS_2 films with the corresponding height profiles by AFM. Clearly, the thickness of the monolayer and bilayer MoS_2 films were ~0.6 and ~1.2 nm, respectively (see the figure, panels **a** and **b**).



▲ **CRITICAL STEP** The commercially available CVD-grown MoS_2 film can be replaced with MOCVD-grown MoS_2 film (see 'Reagents' section for more details). In addition, the thickness of the monolayer and bilayer MoS_2 films are displayed in Box 1.

12. We used a knife to gently scratch the four corners of the baked MoS_2 film, which facilitated an improved reaction between the KOH solution and the SiO_2 beneath the MoS_2 film (Fig. 3a). Supplementary Video 1 provides detailed visual guidance. Subsequently, attach a piece of thermal release tape of appropriate size (~3.5 cm × 4.5 cm) to the baked sample and press and smooth it with a flat plate.
    ▲ **CRITICAL STEP** Thermal release tape was used to facilitate the separation of the MoS_2 film from the Si/SiO_2 substrate. Pressing and smoothing the tape helped to reduce wrinkles and cracks in the MoS_2 film, enabling high-quality transfer.
13. Weigh 14 g of KOH solid-state powder using a medicine spoon, a piece of weighing paper and a balance and then transfer into a glass container. Subsequently, add 125 mL of DI water to prepare a 2 mol/L KOH solution. Next, immerse the sample with the thermal release tape in the KOH solution (Fig. 3b).
14. After ~2 h, use tweezers to gently separate the thermal release tape from the Si/SiO_2 substrate with Supplementary Video 2 provided for visual aid. Then rinse the thermal release tape with the MoS_2 film three times with DI water and put on drying paper to air dry naturally (Fig. 3c).
    ◆ **TROUBLESHOOTING**
15. Attach the air-dried thermal release tape to the Al_2O_3 gate dielectric substrate, pressing it firmly and gently smoothing it with a flat plate.

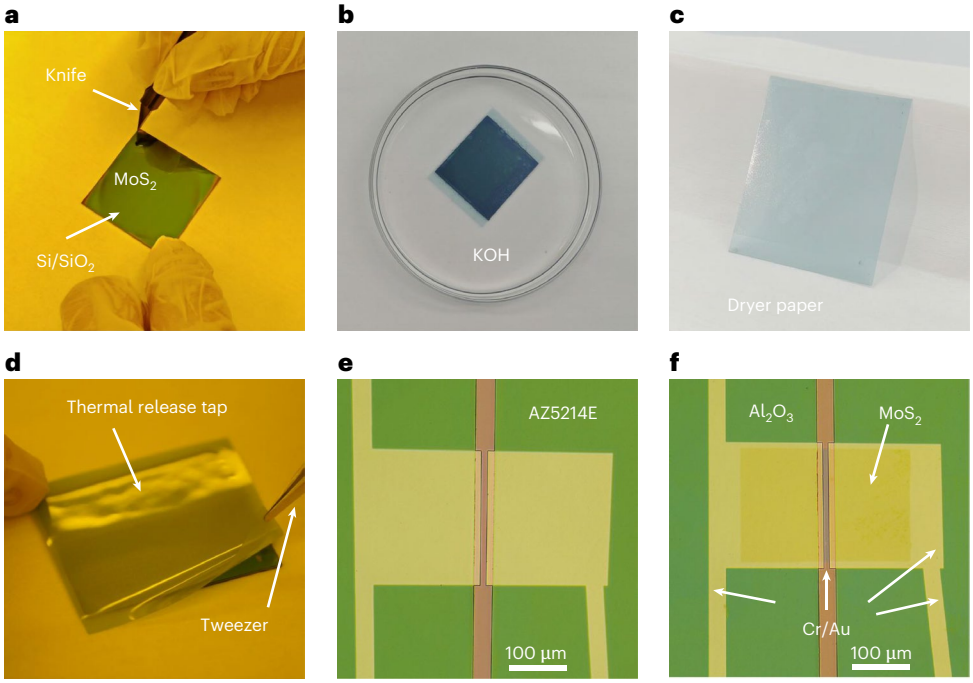


Fig. 3 | The primary process for wet transfer and etching of MoS_2 film. a, An image of the MoS_2 film scraped with a blade. **b**, An image of the sample with the thermal release tape immersed in the KOH solution. **c**, An image of the MoS_2 film dried naturally. **d**, An image of the thermal release tape separated with tweezers. **e**,**f**, OM image before (**e**) and after (**f**) etching the MoS_2 and removing the photoresist.

▲ **CRITICAL STEP** Pressing and smoothing the thermal release tape ensured strong adhesion between the MoS_2 film and the Al_2O_3 gate dielectric, while also reducing wrinkles and cracks in the MoS_2 film. This process was crucial for achieving high-quality transfer.
◆ **TROUBLESHOOTING**

16. Since the thermal release tape changed properties between 120 °C and 140 °C, we set the hot plate to 125 °C. Then, we heated the sample on the hot plate for ~1.5 min. After the heating, we gently peeled off the thermal release tape from one corner using a tweezer, as shown in Fig. 3d. At this point, only MoS_2 and PMMA remained on the gate dielectric substrate.
▲ **CAUTION** Be cautious of burns when using the hot plate. Do not put your hand on its surface.
17. Immerse the sample from Step 16 in acetone for 10 min and repeat the process three times. Then, rinse the sample with IPA for 15 s and dry it with nitrogen gas.
▲ **CRITICAL STEP** It was very critical to repeatedly soak with acetone to ensure the complete removal of PMMA. Use IPA to wash away acetone from the MoS_2 film, but the rinsing time should be brief to prevent wrinkles or damage to the MoS_2 film.
◆ **TROUBLESHOOTING**

Etching 2D semiconductors

● **TIMING 3 h**

18. Spin-coat AZ5214 positive photoresist on the upper surface of the MoS_2 film using Recipe 4 (Table 1), as shown in Fig. 3e.
19. Use laser direct writing or photolithography to pattern the etched window on the baked sample, then develop with AZ300 MIF developer for 20 s, rinse with DI water for 20 s and dry with a nitrogen gun.
20. Use plasma cleaner to etch the MoS_2 film with recipe R5 (Table 1). After etching, immerse the sample in acetone for ~2 h to fully remove the AZ5214 photoresist. As a result, the MoS_2 film was only retained in the channel region (Fig. 3f).

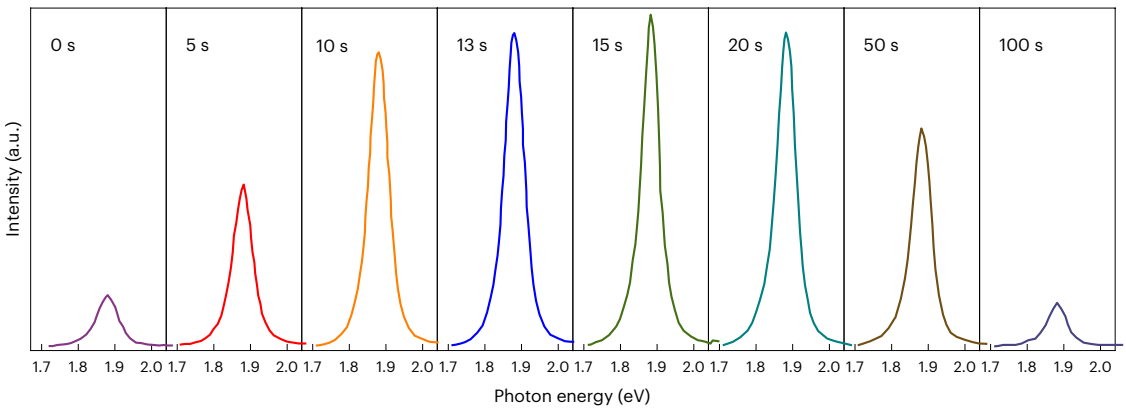
BOX 2

PL of monolayer MoS_2 films with different UVO treatment duration from 0 to 100 s

● **TIMING 11 h**

Procedure

1. Select eight monolayer MoS_2 samples for UVO treatment.
2. Apply UVO treatment to the eight samples using a treatment duration from 0 to 100 s.
3. Considering that the generation of trap states in MoS_2 films during UVO treatment can be precisely characterized by testing the change of PL intensity[4], we used PL to characterize the eight samples, respectively (see the figure). The PL intensity at the peak of 1.88 eV increases with the UVO treatment duration from 0 to 15 s. The UVO treatment introduces more S vacancies in the MoS_2 films, which are active centers for oxygen adsorption. These absorption sites give rise to conversion from trion to exciton emissions and suppress the nonradiative recombination of excitons at defect sites, which leads to the enhancement of PL intensity[4,60]. When the UVO treatment duration exceeds 15 s, the formation of MoO_3—a wide-bandgap indirect semiconductor—leads to a reduction in PL intensity due to the nonradiative recomnation[58]. In this protocol, we aim to introduce more trapping states without oxidizing the MoS_2 film for optimal performance. Therefore, we choose to treat the MoS_2 films using relatively short-time UVO exposure (13 s).



▲ **CAUTION** Before and after using plasma cleaner, it was essential to clean the etching chamber with oxygen plasma to prevent contamination.

◆ **TROUBLESHOOTING**

Annealing

● **TIMING 0.5 h**

21. We annealed the etched sample at 250 °C for 20 min in a rapid thermal annealing processor with a high-purity N_2 environment, which was beneficial for forming good contacts between the contact electrodes and the channel material.

UVO treatment of 2D semiconductors

● **TIMING 0.5 h**

▲ **CRITICAL** UVO treatment of the channel was aimed at increasing the charge-trapping density of the phototransistor, which can promote the hysteresis window, improving the visual adaptation of in-sensor computing devices. This step can be skipped if more charge trapping is unnecessary for the subsequent applications.

22. From the test results in Box 2, it is possible to confirm that UVO treatment can precisely control the generation of trap states. To achieve optimal working performance, we treat the MoS_2 channel with relatively short-time UVO exposure (13 s) at a power intensity of 28 mW/cm^2.

Encapsulation layer preparation

● **TIMING 2 h**

23. As described in Step 7, we used ALD to deposit a 10-nm-thick Al_2O_3 encapsulation layer to protect the phototransistor from performance degradation due to air exposure.

Etching electrode area

● **TIMING 5 h**

24. To expose the source, drain and gate electrodes for electrical contact, it was essential to etch the Al_2O_3 layer covering these electrode regions. Following the procedure introduced in Steps 18 and 19, including spin coating, photolithography, development and drying, we obtained the windows over the electrode regions for etching.
25. Use ICP to etch the Al_2O_3 layer with recipe R6 (Table 1). After etching, dip the sample into acetone for ~2 h to fully remove the AZ5214 photoresist. As a result, the source, drain and gate regions were exposed to air, facilitating subsequent contact with the probe tips.
▲ **CAUTION** It is recommended to always clean the ICP etching chamber before and after use to avoid contamination.

Verification of the devices

Photography and SEM image

● **TIMING 2 h**

26. Place the device prepared with monolayer MoS_2 on a dust-free paper on the experimental table.
27. Take the photograph of the phototransistor array with a camera. The phototransistor array obviously initially verified uniformity with 400 devices and an area of ~3 cm × 3.2 cm (Fig. 4a).
28. Place the prepared device on the stage of the vacuum chamber of the SEM.
29. The length and width of the channel can be obtained by SEM, which were 10 and 200 μm, respectively (Fig. 4b).

Raman spectra and PL spectroscopy

● **TIMING 6 h**

30. Place the monolayer and bilayer MoS_2 films on the stage of a confocal Raman.
31. Test the Raman spectra of the MoS_2 films by the confocal Raman with the excitation laser with a wavelength of 532 nm. To ensure the accuracy of the measurement, we uniformly collected Raman spectra from nine locations across the top, middle and bottom of the sample. The consistent Raman peak positions at ~384.2 and 403.5 cm^{-1}, as well as ~385.3 and 405.6 cm^{-1}, further confirmed that the 2D materials were monolayer and bilayer MoS_2, respectively (Fig. 4c).
32. Place these two MoS_2 films on the stage of the PL measurement system.
33. Characterize the PL spectra of MoS_2 films by using the confocal Raman with a 532-nm laser at an intensity of less than 300 μW. The diffraction grating to collect the PL spectra was 600 grooves per millimeter. Similarly, collect PL spectra from nine dispersed locations. The consistent PL peaks indicated that the bandgap of the monolayer and bilayer MoS_2 were ~1.88 and 1.86 eV, respectively (Fig. 4d).

Optoelectronic characterizations of optoelectronic in-sensor computing devices

Steps 34–58 cover the multitude of characterization tests we routinely carry out on our devices.

Transfer and output curve testing

● **TIMING 1 h**

34. Use the SMU modules of the semiconductor analyzer to test the quasi-direct current electrical characteristics of the device based on this optoelectronic testing system. By sweeping the gate–source voltage (V_G) over a range of ±10 V and testing the channel current at a constant drain–source voltage (V_D), we can get the transfer curves shown in Fig. 5a. The phototransistor exhibited a low off-state current (~1 pA) and a high on/off ratio (~10^8),

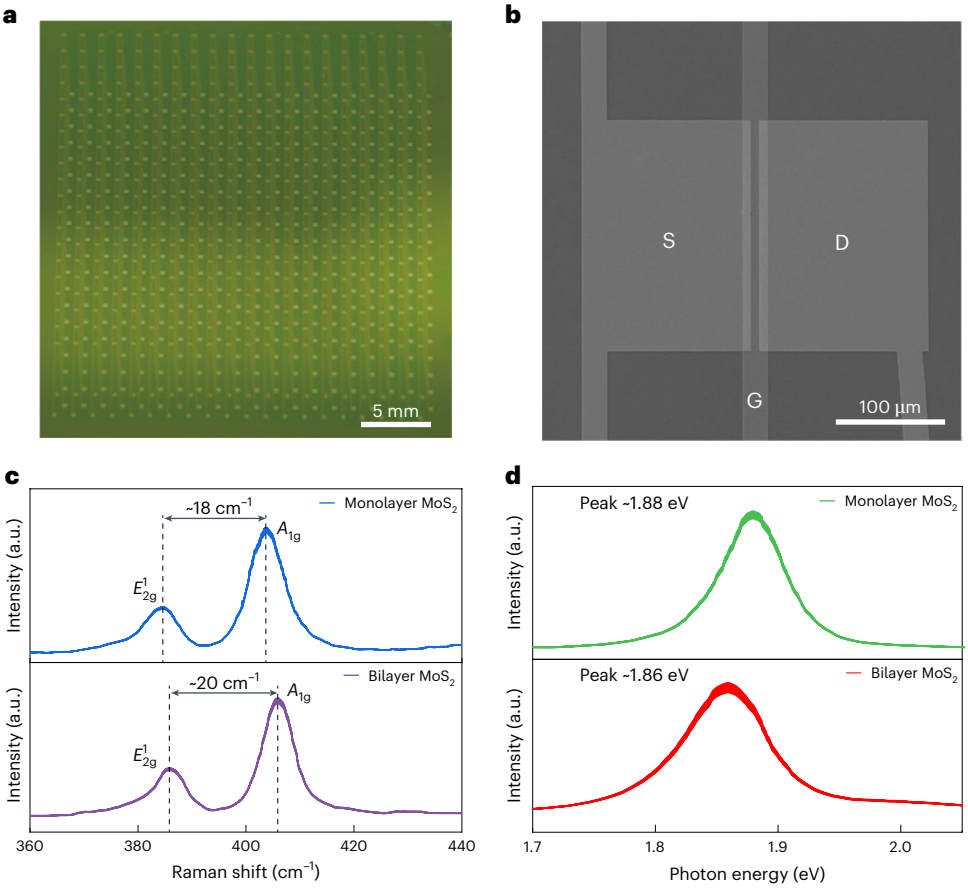


Fig. 4 | Verification of the MoS_2 phototransistors. a, The photography of the phototransistor array. **b**, An SEM image of the device. **c**,**d**, Raman spectra (**c**) and PL spectra (**d**) of the monolayer and bilayer MoS_2 films after device fabrication. A_{1g} and E^1_{2g} correspond to two Raman active modes, namely out-of-plane mode and in-plane mode, respectively. D, drain; G, gate; S, source.

primarily due to high-quality fabrication processes, including gate dielectric growth, channel material transfer and contact electrode deposition.

▲ **CRITICAL STEP** To prevent breakdown of the gate dielectric, the V_G must remain within ±15 V during the measurement of the I_D–V_G transfer curves.

◆ **TROUBLESHOOTING**

35. By sweeping the V_D from 0 to 2 V and testing the channel current at a constant V_G, we obtain the output curves as shown in Fig. 5b. The stable output characteristics indicated good contact between the source–drain electrodes and the channel material, benefiting from optimized annealing and high-quality fabrication techniques.

Charge defect testing

● **TIMING 1 h**

36. To study charge trapping in the bioinspired phototransistor array, we performed double-sweep transfer curves (I_D–V_G) and obtained the hysteresis window (1.2 V) as shown in Fig. 5c. The I_D of the transfer curve increased steeply at the subthreshold region, with a subthreshold swing (SS) value being ~150.2 mV/dec.
37. Fabricate a metal–oxide–semiconductor (MOS) capacitor based on Al_2O_3 with a gate capacitance of ~212 nF/cm^2 (Box 3). Based on the relationship between SS, the trap density (D_{it}) and gate capacitance[55], the D_{it} was calculated to be 1.9×10^{15} cm^{-2}/eV. The large hysteresis window was caused by the charge trapping of MoS_2, which facilitated the phototransistor to obtain more traps, thereby enabling visual adaptation over a wider dynamic range.

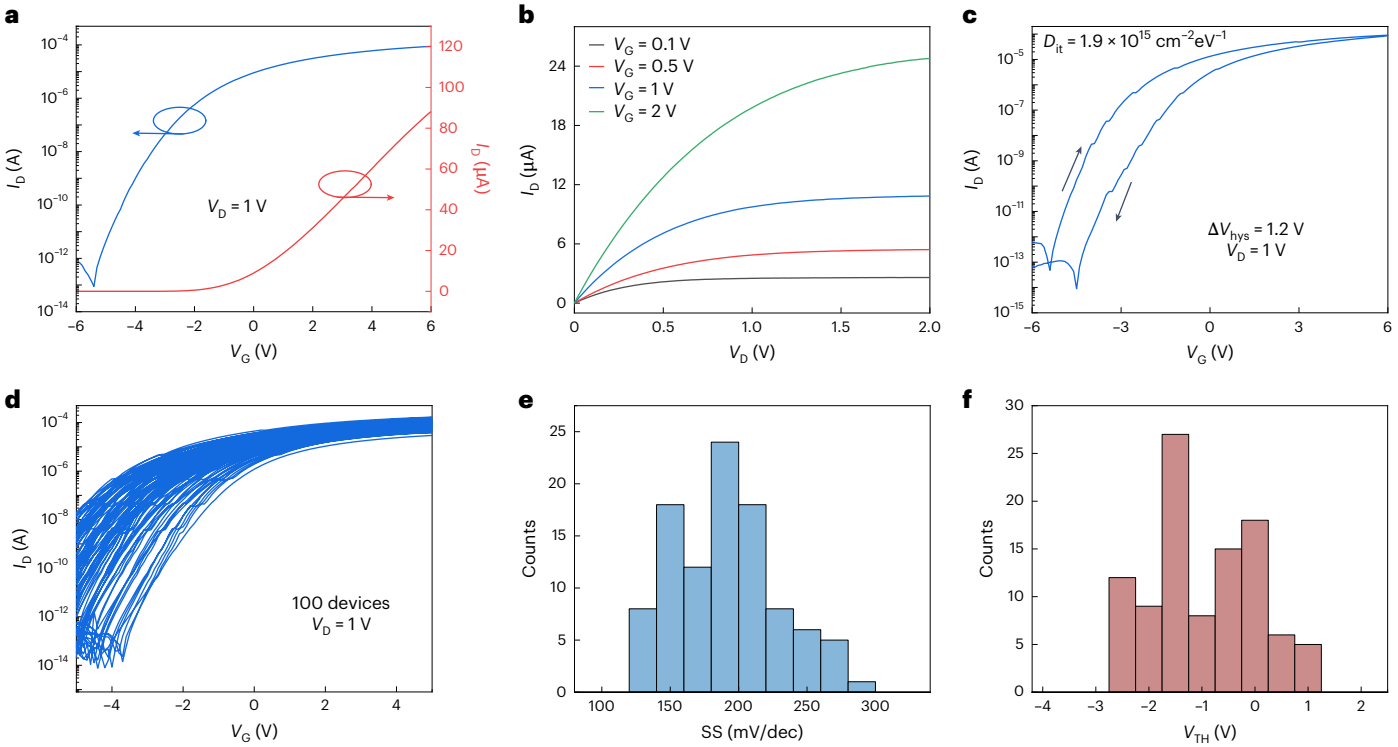


Fig. 5 | Electrical characterization of the MoS_2 phototransistors. a, The transfer curve of a single device tested by changing the V_G range under a dark condition at V_D = 1 V. **b**, The output curves of single phototransistor tested by varying V_D range from 0 to 2 V under a dark condition. **c**, The double-sweep transfer curves of single device at V_D = 1 V in dark conditions. V_{hys}, hysteresis voltage. **d**, The transfer curves of 100 devices under dark conditions at V_D = 1 V. **e**,**f**, The statistical distributions of SS (**e**) and V_{TH} (**f**) of 100 devices under dark conditions.

Uniformity characterization

● **TIMING 12 h**

38. Test the I_D–V_G curves of phototransistors at different locations in the array. Using this method, we sequentially tested 100 randomly selected devices to evaluate the uniformity of the fabricated array.
39. We processed and analyzed the test data from the phototransistor array, yielding the statistical transfer curves presented in Fig. 5d. Based on the analysis methods for SS and threshold voltage (V_{TH})[11,18], we obtained the statistical distributions of SS and V_{TH} for the phototransistor array as presented in Fig. 5e,f. The array devices exhibit SS and V_{TH} values obviously clustered ~200 mV/dec and −1 V, respectively. Compared with state-of-the-art literature on large-area-grown MoS_2 (refs. 11,18), the devices demonstrate comparable uniformity. Furthermore, there are several potential methods to improve uniformity, albeit with greater technical challenges. First, the growth parameters of MoS_2 films can be further optimized, such as substrate selection, temperature control and pressure regulation, to enhance their uniformity. Second, reliable transfer techniques (for example, water-aided transfer[56]) can be developed and optimized to eliminate the contamination arising from the KOH solution etching of the Si/SiO_2 substrate, thereby improving device-to-device uniformity.

Optoelectronic properties at different optical powers

● **TIMING 12 h**

40. According to the optoelectronic testing system, a laser with a wavelength of 660 nm served as the light source, which illuminated the device through a fixed-position optical fiber. Connect a signal generator to the laser to control the illumination pulse, pulse width, pulse period, start time and end time. Use an optical power meter to calibrate the illumination intensity.

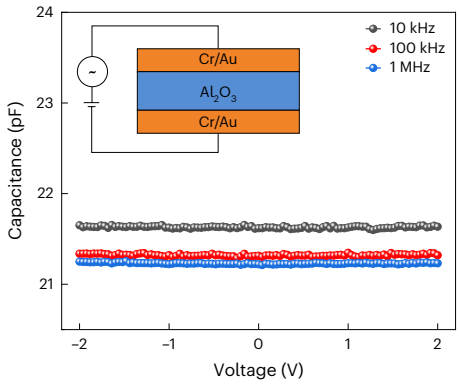
BOX 3

Bias-dependent capacitance density curves of an MOS device

● **TIMING 2 h**

Procedure

1. According to reported capacitance–voltage (*C*–*V*) testing[61], we sequentially deposited 3/30 nm Cr/Au, 40 nm Al_2O_3 and 3/30 nm Cr/Au on Si/SiO_2 substrates to fabricate parallel-plate capacitors, as shown in the figure. Notably, the parallel-plate capacitor had an active area of 1×10^{-4} cm^2.
2. Use the CV modules of the semiconductor analyzer to test the *C*–*V* characteristics of the parallel-plate capacitors and obtain bias-dependent *C*–*V* curves of the metal–oxide–metal device at different frequencies. As the bias voltage swept from −2 to 2 V, the tested capacitance stayed almost stable without fluctuations. Obtain a value of 21.2 pF for gate capacitance at room temperature.



▲ **CRITICAL STEP** The parameters, including light wavelength, light power, the number of pulses, pulse width, pulse period and gate voltage, can be extended on the basis of the structure and materials of the tested in-sensor computing devices.

41. Connect the gate, drain and source electrodes of the device to the probes. We installed an adjustable optical holder onto the optical platform to secure the optical fiber position. Next, we calibrated the laser illumination intensity on the device using a power meter and manual adjustment of the laser's control knob. This enabled testing of I_D–V_G characteristics of the device over the optical power density range from 1 μW/cm^2 to 10 mW/cm^2.
42. Process the test data to obtain the photoresponsivity (*R*) of the devices at different optical powers (Fig. 6a). Considering the influence of low-frequency noise on the detectivity (D^*), we conducted the low frequency noise testing to extract the D^*, which was defined as $D^* = R \times A^{1/2} / S^{1/2}$, where *A* is the active area of the device and *S* is the square of current noise density[57]. Box 4 shows the square of current noise density at different noise frequency under $V_G = 3$ V and $V_D = 1$ V. Given that the current noise density at 1 Hz is usually used for the calculation of D^* (ref. 58), we can get D^* corresponding to various optical powers, as shown in Fig. 6b. These results showed that the fabricated optoelectronic in-sensor computing devices exhibited outstanding optoelectronic properties and could effectively perceive a wide range of light intensities.

Current–time characteristics of the phototransistor

● **TIMING 1 h**

43. To characterize the dynamic stability of the device, we tested the time-varying channel current of the fabricated device at a specific light intensity (1 mW/cm^2) (Fig. 6c), indicating that the device has good stability. The signal generator controlled the laser switching time by adjusting the pulse width. The channel current within a single optical pulse cycle exhibited a fast photoresponse, with a rise time of ~0.92 ms and a fall time of ~10.2 ms (Fig. 6d).

Optoelectronic testing at different wavelengths

● **TIMING 1 d**

44. Based on the optoelectronic measurement system, we replaced the laser with a tunable xenon lamp, which has a wavelength range from 400 nm to 1,000 nm.

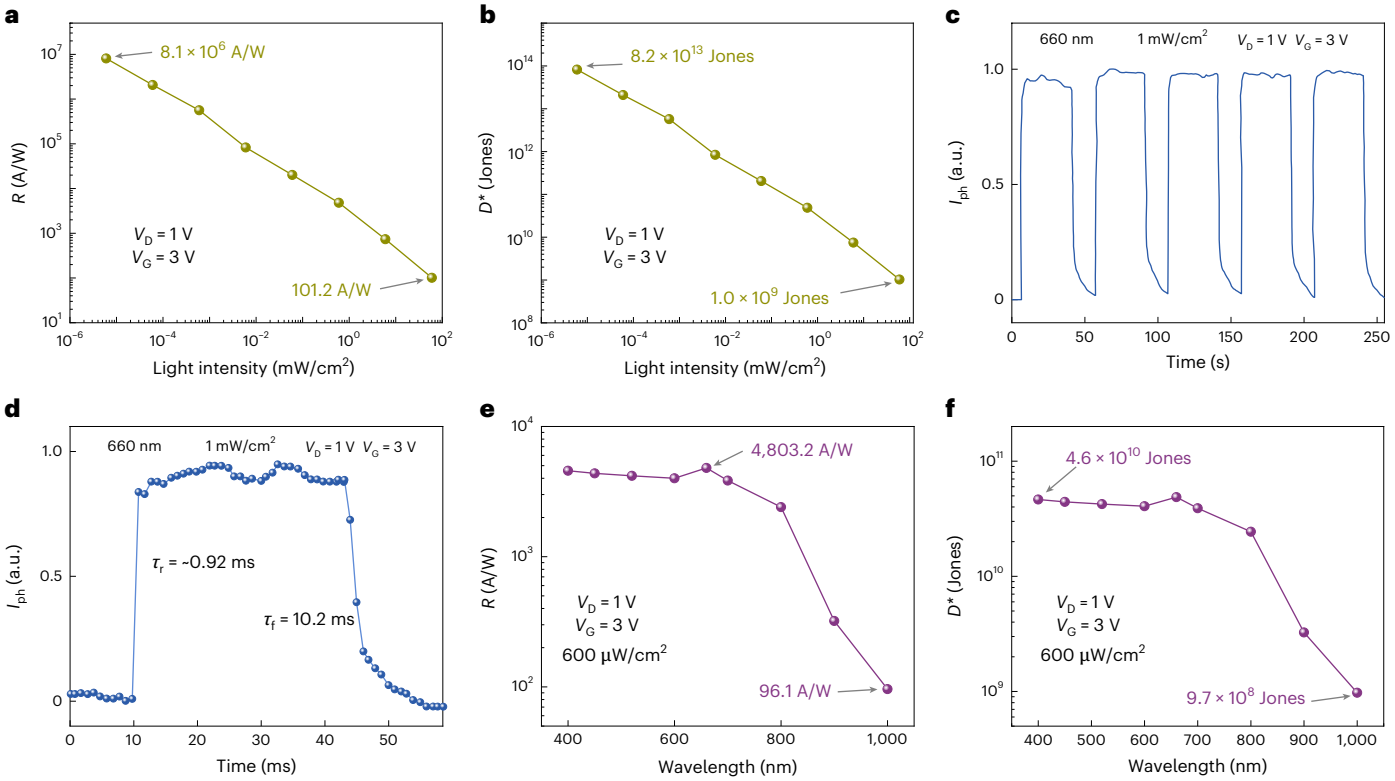


Fig. 6 | Optoelectronic property characterization of the MoS_2 phototransistors. **a**,**b**, R (**a**) and D^* (**b**) of a single device under different illumination intensities. **c**, The time-resolved dynamic measurement result of a single device under the light intensity of 1 mW/cm^2 with V_D = 1 V and V_G = 2 V. a.u., arbitrary unit; I_{ph}, photocurrent. **d**, The rise time and fall time of the device. **e**,**f**, R (**e**) and D^* (**f**) of a single device at various wavelengths with a illumination intensity of 5 mW/cm^2.

45. Before testing, we used an optical power meter to calibrate the optical power of each wavelength to ensure consistency. Then, we tested the I_D–V_G curves of the device sequentially over a wavelength range from 400 to 1,000 nm. By processing test data with consideration of low-frequency noise (Box 4), we obtained the R and D^* of the fabricated devices at various wavelengths (Fig. 6e,f). The MoS_2 treated with UVO contains several localized trap states within the band gap, primarily due to the trap configurations of sulfur vacancies[59]. These traps form new energy levels that can absorb light within a specific wavelength range, thereby broadening the detection spectrum of the channel. In addition, the introduction of charge trapping can prolong the recombination time of photogenerated electron–hole pairs, allowing the carriers to remain in the channel longer, thereby improving the R. Furthermore, high-quality fabrication can reduce the contact barrier between the channel material and the metal electrode, enabling more photogenerated carriers to reach the electrodes and generate photocurrent, thereby increasing the R and D^*. Consequently, these factors enable the in-sensor device to be applied in visual perception in a wide wavelength range.

PR of this phototransistor

● **TIMING 2 h**

46. Based on the optoelectronic testing system, we tested the transfer curves (I_D–V_G) of devices under the dark condition and illumination at 660 nm. Adjust the light intensity with an optical power meter, ranging from 6 nW/cm^2 to 60 mW/cm^2. For the I_D–V_G curves, set the V_D to 1 V and sweep the V_G from −6 V to 6 V with a step of 0.05 V.

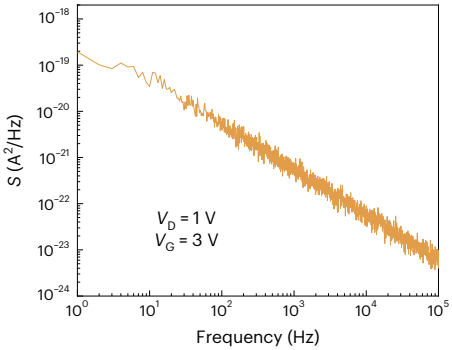
BOX 4

Current noise of the phototransistor as a function of frequency

● **TIMING 2 h**

Procedure

1. Conduct low-frequency noise tests of the phototransistor using the specialized noise analysis module of a semiconductor parameter analyzer (Platform Design Automation, FS-Pro). During testing, connect the source, drain and gate electrodes of the device to three probes the semiconductor analyzer.
2. Set the gate voltage and bias voltage to 3 V and 1 V, respectively. Obtain the current noise of the phototransistor by sweeping the frequency from 1 to 10^5 Hz.



47. When the light intensity was 60 mW/cm^2, according to the relationship between I_D and effective PR[4], the calculated PR of the phototransistor reached 170 dB at 6 V (Fig. 7a). This indicated that the fabricated devices were capable of visual perception over a wide illumination range.

Adaption characterization

● **TIMING 3 h**

48. As described in Steps 40 and 43, we tested the I_D–t curves of the device under different V_G according to different illumination intensities.
49. Test the I_D–t curves under continuous weak light of 5 μW/cm^2 and dim background illumination at 500 nW/cm^2 by applying V_G of −4 V. Based on the relationship between I_D and current change ratio (CCR)[4], processing the tested data yielded the CCR values of 26.9 (Fig. 7b). For comparison, test I_D–t curves under continuous strong light of 5 mW/cm^2 and bright background illumination at 500 μW/cm^2 by applying a V_G of +6 V. An analysis of the data yielded CCR values of 0.73 (Fig. 7c). These results indicated that V_G can modulate the time-varying excitation or inhibition characteristics of the in-sensor device. Specifically, applying a more negative V_G under dim backgrounds enhanced the excitation characteristics, while applying a more positive V_G under bright light led to inhibition behavior, thereby realizing visual adaptation (both scotopic and photopic adaptation).

Visual adaptation

● **TIMING 1 d**

50. Select an 8 × 8 pixel array from the MoS_2 phototransistor array to perceive the pattern of 'P' (Fig. 1). Following the procedures described in Steps 34, 37 and 40, we sequentially connected three probes to the gate, drain and source electrodes of each phototransistor in the arrays to test their I_D–t curves. Notably, all devices within each column shared a common source and gate, which greatly reduced the need to reposition the probes for the source and gate.
51. For the scotopic adaptation testing of the MoS_2 phototransistor array, 15 devices corresponding to the image 'P' pattern were under a weak light of 5 μW/cm^2, while the other 49 devices were under a dim background illumination at 500 nW/cm^2 (Fig. 1). Apply a gate voltage of −4 V to all devices and extract the I_D at a different time from the corresponding

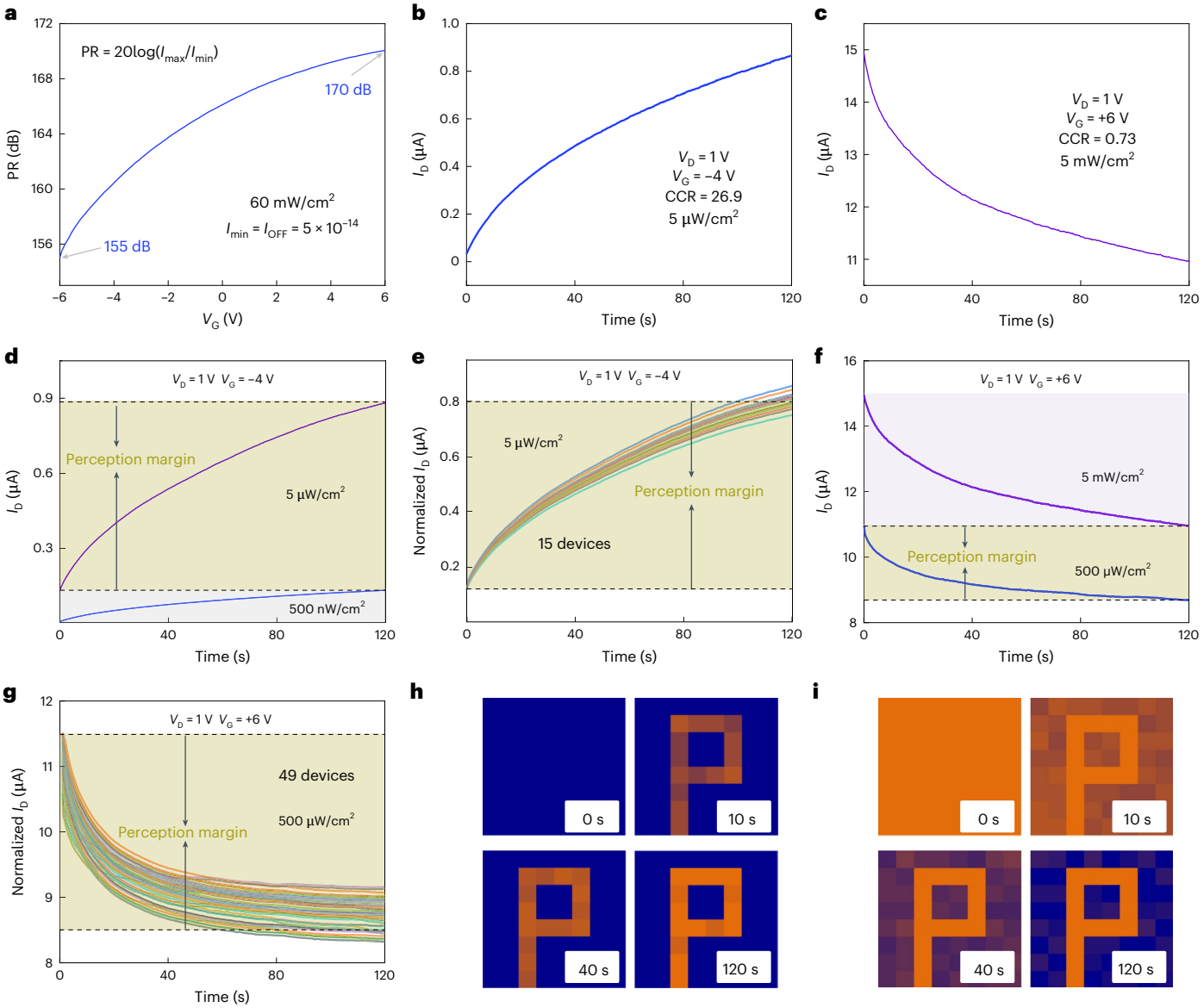


Fig. 7 | Bioinspired in-sensor computing characterization for visual adaptation. a, The effective PR dependent on V_G. I_{max} is I_D of the device under 60 mW/cm², while both I_{min} and I_{OFF} represent off-state current of the device. **b**, The time-dependent I_D of the in-sensor device under continuous illumination of 5 μW/cm² at V_G = −4 V. **c**, The time-dependent I_D of the device under continuous illumination of 5 mW/cm² at V_G = +6 V. **d**, The time-dependent I_D of the device under 5 μW/cm² and 500 nW/cm² at V_G = −4 V before the I_D normalized. **e**, The time-dependent normalized I_D of 15 devices under 5 μW/cm² at V_G = −4 V. **f**, The time-dependent I_D of the device under 5 mW/cm² and 500 μW/cm² at V_G = +6 V before the I_D normalized. **g**, The time-dependent normalized I_D of 49 devices under 500 μW/cmv at V_G=+6 V. **h**,**i**, The time process of scotopic (**h**) and photopic (**i**) adaptation for the image 'P' pattern.

I_D–t curves. Obtain the image contrast (C) by calculating the difference in the gray level (G) between the two kinds of pixels inside 'P' and outside 'P' (ref. 4). Taking the I_D–t curves obtained from two devices tested under weak light and background illumination as an example, to improve the adaptation effect, it was essential to consider the perception margin during the scotopic adaptation process to improve the adaptation function (Fig. 7d). To maintain the perception consistency of the array device, we defined I_D = 0.12 μA and 0.8 μA corresponding to G of 0 and 255. The definition method guaranteed that the light response G remained at 0 under background illumination, allowing the extraction of the normalized I_D variations over time from 15 devices under a weak light (Fig. 7e).

Consequently, the *C* of the image changed only with the I_D of these 15 devices, thereby enabling the scotopic adaptation process over time (Fig. 7h).

52. Similarly, for the photopic adaptation testing, 15 devices corresponding to the pattern of 'P' were under a strong light of 5 mW/cm^2, while the other devices were under bright background illumination at 500 μW/cm^2. Apply a gate voltage of +6 V to all devices and extract the I_D at different time from the corresponding I_D–*t* curves. Then, we can obtain the perception margin under strong light and bright background illumination, as well as the normalized I_D of 49 devices changing over time under bright background illumination (Fig. 7f,g). The *C* of the image changed only with the I_D of these 49 devices, illustrating the photopic adaptation process over time (Fig. 7i). The longer the device perceived the image, the more accurate the image recognition became. The scotopic and photopic adaptation processes indicated that the optoelectronic in-sensor computing devices exhibited computational capability for feature enhancement, which can simplify the complexity of hardware and algorithms and enhance the image processing capabilities at sensory terminals.

Grade neuron property of this phototransistor

● **TIMING 2 h**

53. As described in Step 40, we adjusted the output pulse width of the signal generator to set the stimulation frequency of the 660 nm laser to 100 Hz. The laser intensity was 4 mW/cm^2.
54. Adjust the number of output pulses from the signal generator (for example, four) and test the I_D–*t* curve. Owing to the pulse duration being shorter than the recovery time of the phototransistor, the device can emulate the grade neuron characteristics through cumulative photocurrent under multiple consecutive pulses (Fig. 8a).
55. By changing the number of pulses, the I_D gradually increased and tended to saturate (Fig. 8b). The change trend was because the carriers trapped after light illumination were incompletely released within the pulse interval. Subsequent light irradiation generated more carriers and filled more trap states, resulting in higher I_D. This nonlinear increase in I_D mimicked the response of grade neuron characteristics.

Time constant modulated by gate voltage

● **TIMING 6 h**

56. The MoS_2 phototransistors have shallow and deep trap states. The V_G of phototransistors can modulate the trapping/detrapping characteristics of the channel, thereby affecting the photoresponse and falling time of the device. By testing the I_D–*t* curves at V_G of −3 V, 0 V and 3 V with V_D = 0.1 V, we can obtain the time constant for each gate voltage (Fig. 8c), which were 10, 10^2 and 10^6 ms, respectively. These results showed that the in-sensor computing devices can perceive a wide range of motion scenes at sensory terminals.

Motion perception

● **TIMING 1 d**

57. Based on the optoelectronic testing system, the light source (660 nm) first projected a pentagram pattern through the mask onto a 20 × 20 MoS_2 phototransistor array. As described in Step 50, we sequentially connected the gate, drain and source electrodes of each phototransistor in the array to probes to test their I_D–*t* curves. Similarly, all devices in each column shared a common source and gate electrode.
58. For the motion perception testing of the phototransistor array, we used four temporal frames to represent the output of the light source with 4-mW/cm^2 intensity and 100-ms pulse width. Then, we projected the pentagram pattern onto the array devices in four motions with a speed of 0.5 m/s: moving right, moving left, approaching and leaving. With a gate voltage of V_G = 0 V applied to all devices, we can obtain their corresponding I_D–*t* curves by a sampling frequency of 10 Hz (ref. 5). Considering that the photocurrent of the in-sensor computing device gradually increased (continuous light stimulation) or decayed (after light stimulation) over time, the summation of the photocurrent frames can clearly show the contour of the entire motion trajectory of the object, including rightward, leftward, approaching and leaving (Fig. 8d).

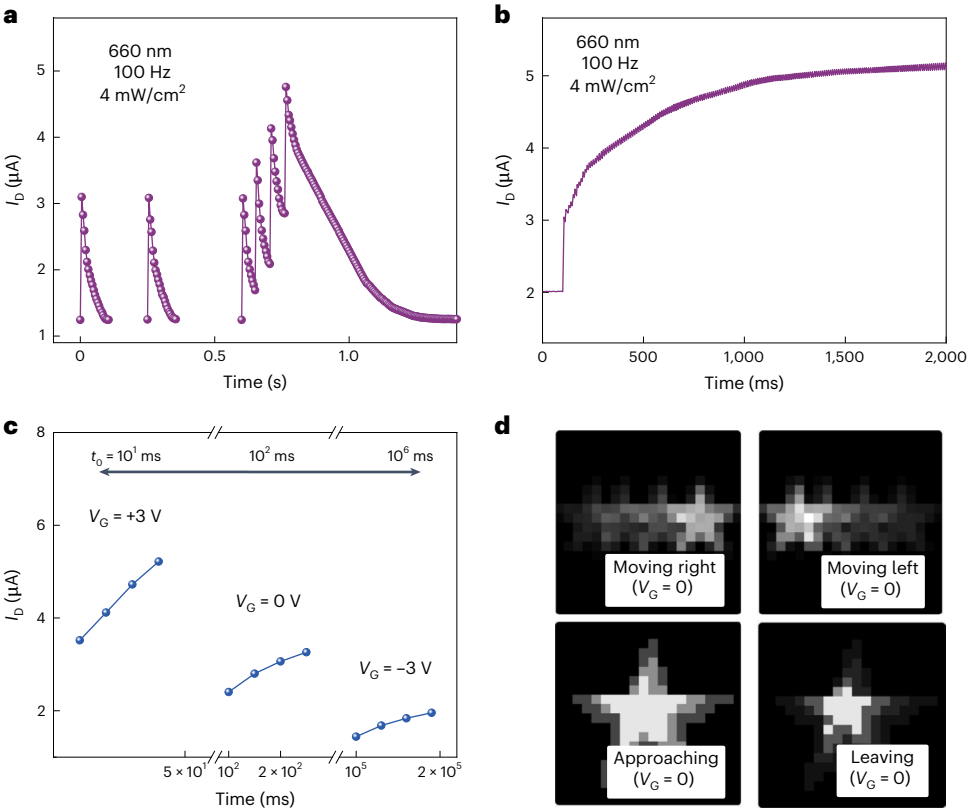


Fig. 8 | Bioinspired in-sensor computing characterization for motion perception. a, The time-dependent I_D under a single and 4-bit light pulse stimulation. **b**, The time-dependent I_D of the in-sensor computing device for a stimulation frequency of 100 Hz. **c**, The time constants of the device varying from 10^1 ms to 10^6 ms under different V_G. **d**, The output current maps of the phototransistor array for four motions (moving right, moving left, approaching and leaving) at $V_G = 0$ V.

Troubleshooting

Troubleshooting advice can be found in Table 2.

Table 2 | Troubleshooting table

Step	Problem	Possible reason	Solution
5	Many metals cannot be successfully lifted off when using single-layer photoresist (AZ5214E)	During liftoff, acetone has difficulty penetrating the single-layer photoresist beneath the metal	Use a dual-layer photoresist (PMGI SF6 and AZ5214E) for both lithography and liftoff processes
7	The grown Al_2O_3 is loose and exhibits a high roughness	Relatively low growth temperature	Use relatively high temperature (~290 °C) to grow Al_2O_3
14	The thermal release tape is difficult to separate from the substrate	The sample is immersed in the KOH solution for a relatively short time	Before separation, the sample is immersed in the KOH solution for ~2 h
15	Slipping occurs when pressing the thermal release tape	The thermal release tape was not fully air-dried	Extend the natural drying time or dry it gently with a nitrogen gun
17	The MoS_2 film at the edge of the contact electrode is easy to break	The deposited electrodes are relatively thick, resulting in high stress at the edges	Deposit thinner Cr/Au (3/30 nm) contact electrodes
20	After etching, the photoresist used as a barrier layer is difficult to remove	O_2 plasma can react with the photoresist, leading to its degradation	Optimize etching time (~50 s)
34	The *I*–*V* curve suddenly changes sharply during the testing	These probes do not establish good contact with the device	Before testing, reposition the probes to achieve proper contact with the device

Timing

Steps 1–5, back gate and metal marker preparation: 1 d
Steps 6–8, gate dielectric and contact electrode preparation: 1–2 d
Steps 9–17, wet transfer of 2D semiconductors: 1 d
Steps 18–20, etching 2D semiconductors: 3 h
Steps 21–22, UVO treatment of 2D semiconductors: 1 h
Steps 23–25, encapsulation layer preparation and etching electrode area: 1 d
Steps 26–33, verification of the devices: 1 d
Steps 34–45, optoelectronic characterizations of optoelectronic in-sensor computing devices: 2–4 d
Steps 46–58, bioinspired in-sensor computing characterization: 4 d

Anticipated results

For optoelectronic applications of 2D semiconductors, the large dynamic range of the photodetector facilitates low-light detection. The UVO treatment of the channel is expected to increase the hysteresis window of the phototransistor by introducing more trap centers and trap states and increasing D_{it}, which enhances the photogating effect and results in an improved dynamic range for the photodetector. The double-sweep transfer curves of the device are depicted in Fig. 9a with and without the UVO treatment at $V_D = 1$ V in the dark. Obviously, UVO treatment can increase the hysteresis window of the phototransistor. In addition, the reproducibility of optoelectronic performance of fabricated devices may be the most critical challenge. In this protocol, we performed a sampling study to compare the yield of MoS_2 with varying growth methods and batches. Phototransistors with identical structures were fabricated using monolayer MoS_2 grown by CVD and bilayer MoS_2 grown by MOCVD. By testing the I–V characteristic curves of phototransistor arrays from three batches grown using different methods, the statistical results were obtained, as shown in Fig. 9b. With the continued advancement of growth techniques, the devices fabricated by both growth methods exhibit high yield. The yield of photodetectors fabricated in different batches using the same growth method show variations, indicating that the reproducibility of the growth method still requires further optimization. Further, the yield and repeatability of the phototransistor based on the monolayer MoS_2 are slightly lower, possibly owing to the thinner nature of monolayer MoS_2, which makes it more susceptible to damage during growth or transfer. In addition, the thinner MoS_2 layers are more vulnerable to damage during E-beam deposition of the contact electrodes. This compromises the contact quality between the source–drain electrodes and the channel, leading to lower R in phototransistors based on monolayer MoS_2 (Fig. 9c).

In summary, we propose that phototransistors based on monolayer MoS_2 offer a suitable choice for future optoelectronic applications owing to their optimal balance between fabrication efforts and optoelectronic performance. Moreover, 4-inch MoS_2 films can already be grown in the laboratory, which is just the beginning. As growth techniques continue to improve, we are optimistic that MoS_2 films with larger areas, greater uniformity and enhanced performance will be developed. In addition, the whole procedure described in this protocol, including electrode deposition, gate dielectric growth, etching, laser cutting, annealing and even transferring MoS_2 to the target substrate, can be industrialized, scaled up and made intelligent. Furthermore, MoS_2 film is a representative example, and a wide range of semiconducting materials can be used to fabricate in-sensor computing devices. Considering the low power consumption and simplicity of optoelectronic in-sensor computing devices and circuits, we believe that in-sensor computing devices hold significant potential for applications in machine vision systems.

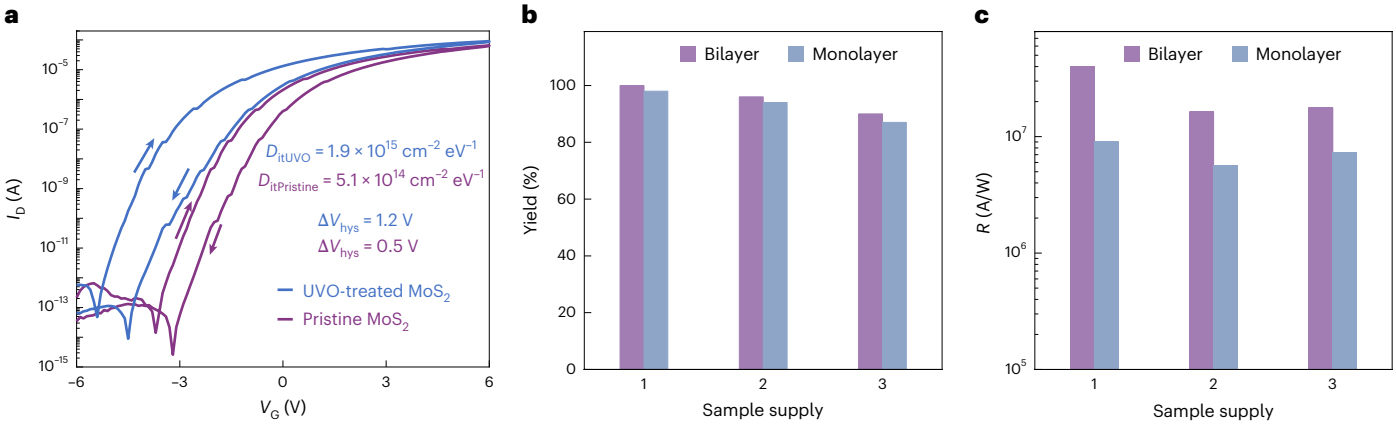


Fig. 9 | Transfer curves and parameter comparison of different MoS_2 phototransistors. a, The double-sweep transfer curves of a single device with and without the UVO treatment at $V_D = 1$ V in the dark condition. D_{itUVO} and $D_{itPristine}$ refer to the trap density of the device with and without the UVO treatment, respectively. **b**,**c**, The yield (**b**) and *R* comparison (**c**) of phototransistors from three batches of monolayer and bilayer MoS_2 film grown using CVD and MOCVD.

Data availability

The main data supporting this protocol are available in our previous publications[4,5] and can be obtained from the corresponding author upon reasonable request. Source data are provided with this paper.

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Author contributions

Y.C. conceived the idea. G.Z., S.M. and H.C. developed the protocol and performed the fabrication and testing. G.Z., S.M., T.W. and J.C. compiled the experimental data. G.Z., T.W. and H.C. designed video supplements. G.Z. and Y.C. cowrote the manuscript. All the authors discussed the results and contributed to writing this manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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